ic_autolab round 28 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T08:36:02Z at git `aeb70e6d36bf02ae6a7e63b657382edff831122c`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-d87806` (`python3 tools/pcc/pcc.py standing PCC-20261104-d87806`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
20	**improvement**	wsum2g-b15-n2	0.515 [0.474, 0.559]	0.513 [0.470, 0.560]	0.499 [0.441, 0.566]	sel < inc	2.575 / 2.691 / 3.741	5.001 / 5.249 / 7.489	1.463 / 1.565 / 2.490	1.8x / 1.7x / 1.5x
32	**no_challenger**	none (best candidate `wsum2g-b120-n2`, train S 2.880 / 3.042 / 4.503)	-	-	-	-	-	2.561 / 2.809 / 5.041	1.047 / 1.150 / 2.077	inc: 2.4x / 2.4x / 2.4x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

- k = 20: compare ops.status = ok, admissible = true; sel / inc = 0.5149 [0.474, 0.559] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 32: no_challenger

Train (development screen, in sample)

k = 20: session `ECBS1hfb1ef1a16bb8` (run exit 0, verify exit 0, receipt sha256 `0d991304e841507cbcf04d7bd9cd62933cdbc2c1c7ea0128628a76f39c657b04`)

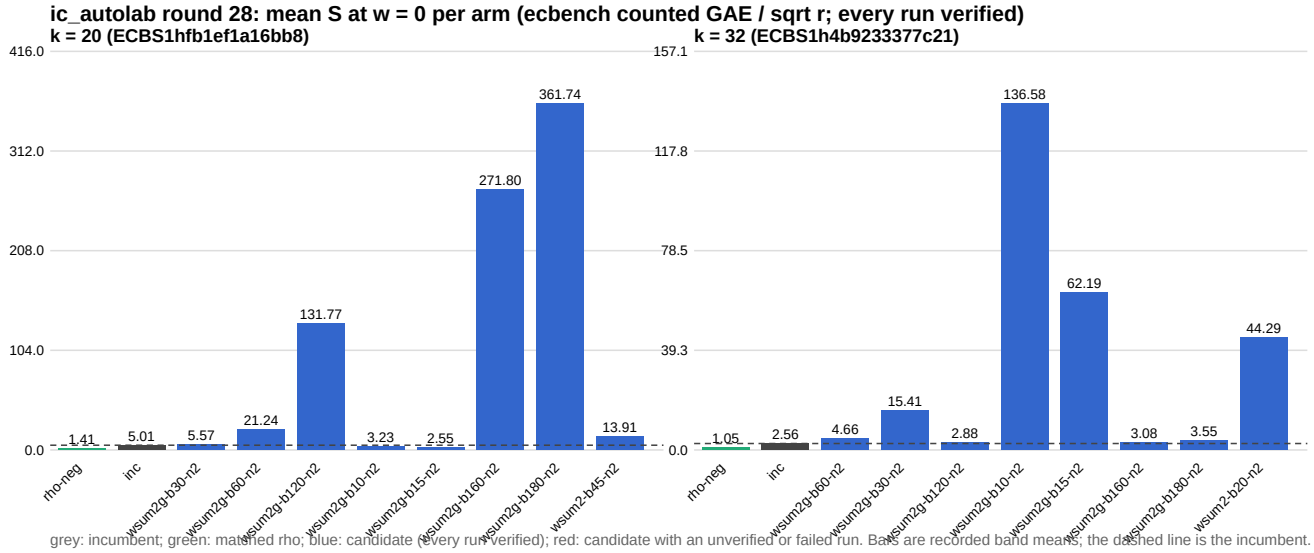
arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	`--`	`rho.negation`	32	32	1.407 / 1.505 / 2.383	matched reference
`inc`	wsum2-b15	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=15,free=1`	32	32	5.009 / 5.263 / 7.552	incumbent
`wsum2g-b30-n2`	wsum2g-b30	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=30,free=1`	32	32	5.575 / 5.611 / 5.934	cross-size transfer: wsum2g-b30-n2 is the k = 24 incumbent at N / N_a = 0.025
`wsum2g-b60-n2`	wsum2g-b60	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=60,free=1`	32	32	21.241 / 21.375 / 22.585	cross-size transfer: wsum2g-b60-n2 is the k = 28 incumbent at N / N_a = 0.010
`wsum2g-b120-n2`	wsum2g-b120	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=120,free=1`	32	32	131.772 / 139.606 / 210.110	exploration: family wsum2g-b120 untried at k = 20, N = 0.0001 x N_a

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`wsum2g-b10-n2`	wsum2g-b10	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=10,free=1`	32	32	3.228 / 3.483 / 5.771	exploration: family wsum2g-b10 untried at k = 20, N = 0.0001 x N_a
`wsum2g-b15-n2`	wsum2g-b15	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=15,free=1`	32	32	2.553 / 2.667 / 3.698	exploration: family wsum2g-b15 untried at k = 20, N = 0.0001 x N_a
`wsum2g-b160-n2`	wsum2g-b160	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=160,free=1`	32	32	271.796 / 291.391 / 467.741	exploration: family wsum2g-b160 untried at k = 20, N = 0.0001 x N_a
`wsum2g-b180-n2`	wsum2g-b180	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=180,free=1`	32	32	361.741 / 389.203 / 636.366	exploration: family wsum2g-b180 untried at k = 20, N = 0.0001 x N_a
`wsum2-b45-n2`	wsum2-b45	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=45,free=1`	32	32	13.915 / 13.953 / 14.297	exploration: family wsum2-b45 untried at k = 20, N = 0.0001 x N_a

k = 32: session `ECBS1h4b9233377c21` (run exit 0, verify exit 0, receipt sha256 `163c7f277f2147025fd2bfc23b00ebe2907dafa5a145a2bc2348a84ace5850d9`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	`--`	`rho.negation`	32	32	1.047 / 1.150 / 2.077	matched reference
`inc`	wsum2-b120	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=120,free=1`	32	32	2.561 / 2.809 / 5.041	incumbent
`wsum2g-b60-n2`	wsum2g-b60	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=60,free=1`	32	32	4.655 / 5.089 / 8.991	cross-size transfer: wsum2g-b60-n2 is the k = 28 incumbent at N / N_a = 0.010
`wsum2g-b30-n2`	wsum2g-b30	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=30,free=1`	32	32	15.413 / 16.946 / 30.742	cross-size transfer: wsum2g-b30-n2 is the k = 24 incumbent at N / N_a = 0.025
`wsum2g-b120-n2`	wsum2g-b120	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=120,free=1`	32	32	2.880 / 3.042 / 4.503	exploration: family wsum2g-b120 untried at k = 32, N = 0.0001 x N_a
`wsum2g-b10-n2`	wsum2g-b10	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=10,free=1`	32	32	136.580 / 150.236 / 273.146	exploration: family wsum2g-b10

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
wsum2g-b15-n2	wsum2g-b15	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=15,free=1	32	32	62.187 / 68.403 / 124.349	exploration: family wsum2g-b15 untried at k = 32, N = 0.0001 x N_a
wsum2g-b160-n2	wsum2g-b160	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=160,free=1	32	32	3.075 / 3.160 / 3.924	exploration: family wsum2g-b160 untried at k = 32, N = 0.0001 x N_a
wsum2g-b180-n2	wsum2g-b180	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=180,free=1	32	32	3.546 / 3.619 / 4.276	exploration: family wsum2g-b180 untried at k = 32, N = 0.0001 x N_a
wsum2-b20-n2	wsum2-b20	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=20,free=1	32	32	44.295 / 53.117 / 132.521	exploration: family wsum2-b20 untried at k = 32, N = 0.0001 x N_a



train screen

Selection

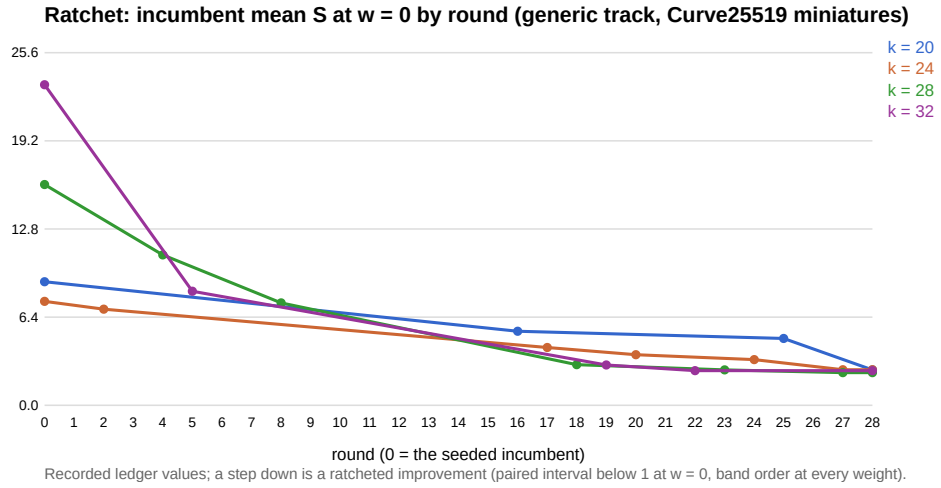
- k = 20: challenger `wsum2g-b15-n2` (Pareto alternative at w = 1: `wsum2g-b15-n2`)
- k = 32: no_challenger (Pareto alternative at w = 1: `wsum2g-b160-n2`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
20	ECBS1hedb8cab33622	0	0	d1fb88594a6510e05d9d954c155267d2ddf2d53679834041230a703bb5374c71	{“verified”:480}

Ledger after this round

k	incumbent	since round S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum2g-b15-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=15,free=1`)	28 2.575 / 2.691 / 3.741	1.8x / 1.7x / 1.5x
24	`wsum2g-b30-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=30,free=1`)	27 2.584 / 2.711 / 3.855	2.0x / 2.0x / 1.6x
28	`wsum2g-b60-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=60,free=1`)	27 2.373 / 2.483 / 3.472	2.4x / 2.3x / 1.8x
32	`wsum2-b120-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=120,free=1`)	22 2.519 / 2.759 / 4.915	2.7x / 2.7x / 2.7x



ratchet

Evidence and deviations

- ecbench binary sha256: `6229f85b81f9a2e8f2811d675f77f4f830ca536422a1e92680a382183450a62c`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.